Late-Layer Feature Reweighting by an Alignment Adapter: Improving SAE Quality with Muon

Abstract

We study where a constitutional alignment adapter acts inside a small language model using a reproducible mechanistic interpretability pipeline built around balanced scenario sampling, layerwise activation capture, sparse autoencoder (SAE) training, fixed-base drift analysis, and causal feature steering. Across the run artifacts available here, the strongest causal evidence comes from held-out layer-23 steering: the target subspace $\{212, 714\}$ shows a positive dose-response on authored-target likelihood, while a matched control subspace $\{978, 193\}$ does not. The best held-out dose for the target subspace is coefficient -0.75, with mean $\Delta \log p = +0.2019$ and mean likelihood ratio 1.2431. These results support a view of the constitutional adapter as a late-layer feature reweighting mechanism rather than a broad distributed rewrite.

1 Core claim

The strongest defensible claim from the current evidence is:

> On this benchmark, the constitutional adapter acts primarily by reweighting a narrow layer-23 reasoning subspace rather than by broad distributed changes across the network.

Formally, we analyze intervention effects using target-likelihood deltas:

$$\Delta \log p(y^* \mid x; \alpha, S) = \log p(y^* \mid x; \alpha, S) - \log p(y^* \mid x; 0, S),$$

where y^* is the authored target continuation, x is the prompt, S is the steered SAE feature set, and α is the steering coefficient.

2 Optimizer benchmark results

The SAE optimizer bakeoff establishes the practical training recipe used in the project. Muon-hybrid improves quality-sensitive metrics on Windows CUDA, while WSL Flash-Muon improves reconstruction but loses badly on efficiency.

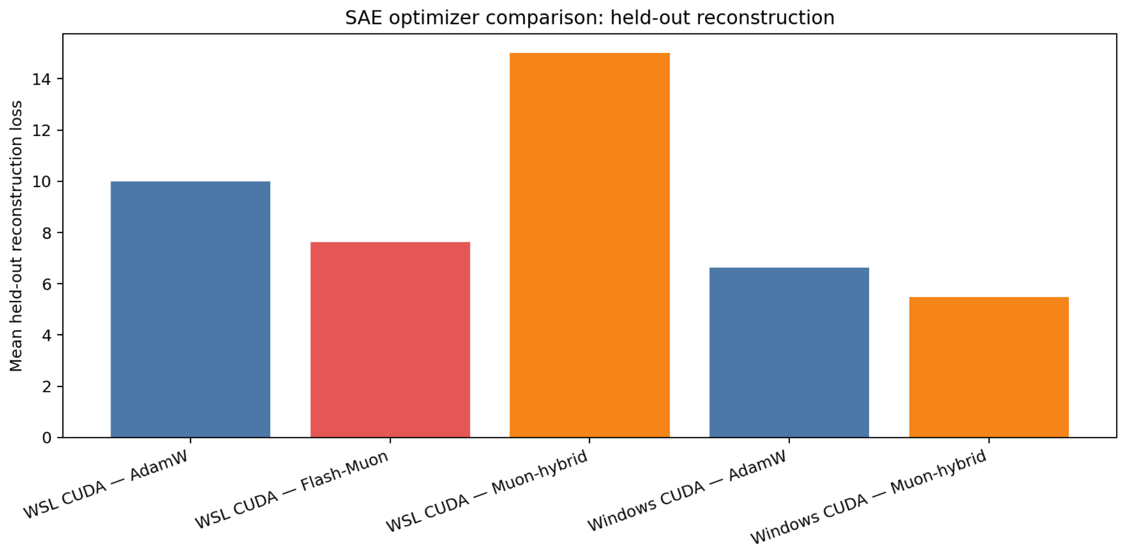

Figure 1: SAE optimizer comparison: mean held-out reconstruction loss across the available Windows and WSL benchmark runs. Lower is better.

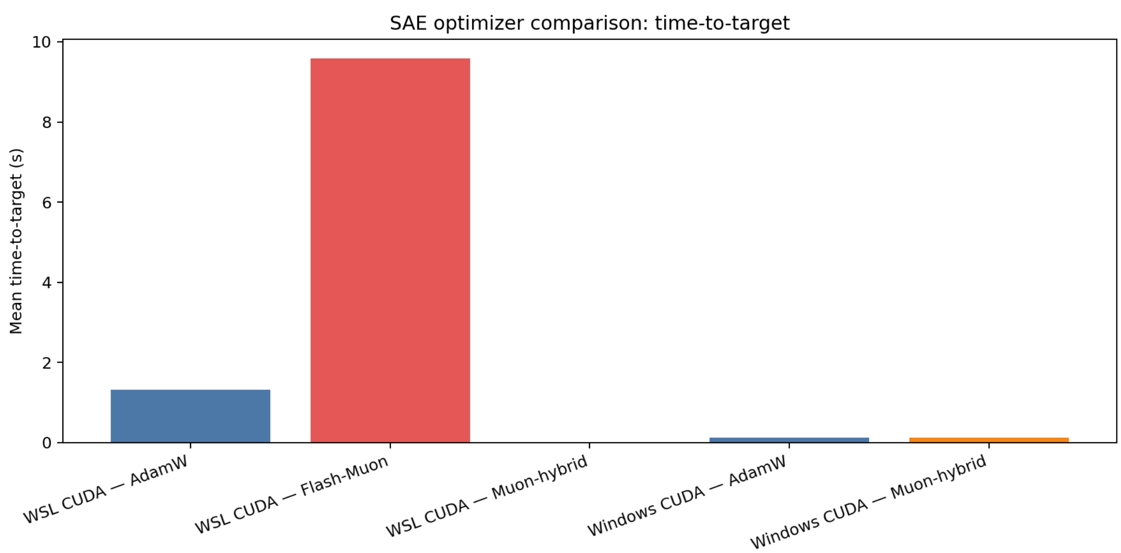

Figure 2: SAE optimizer comparison: mean time-to-target across the available benchmark runs. Lower is better.

Interpretation:

- **Windows Muon-hybrid** is the best practical quality/efficiency tradeoff.
- **Windows AdamW** remains the throughput baseline.
- **WSL Flash-Muon** is usable but not worth adopting on this stack for the main pipeline.

3 Causal steering results

The final causal experiment uses held-out layer-23 joint steering with matched controls:

- target set: $\{212, 714\}$
- control set: $\{978, 193\}$
- world: `mq_constitution_floodplain_v2`
- prompts: `page_a1_triage`, `page_a2_router`, `page_a3_truth`
- intervention: last-token-only residual steering
- metric: authored `final_output_preview` likelihood

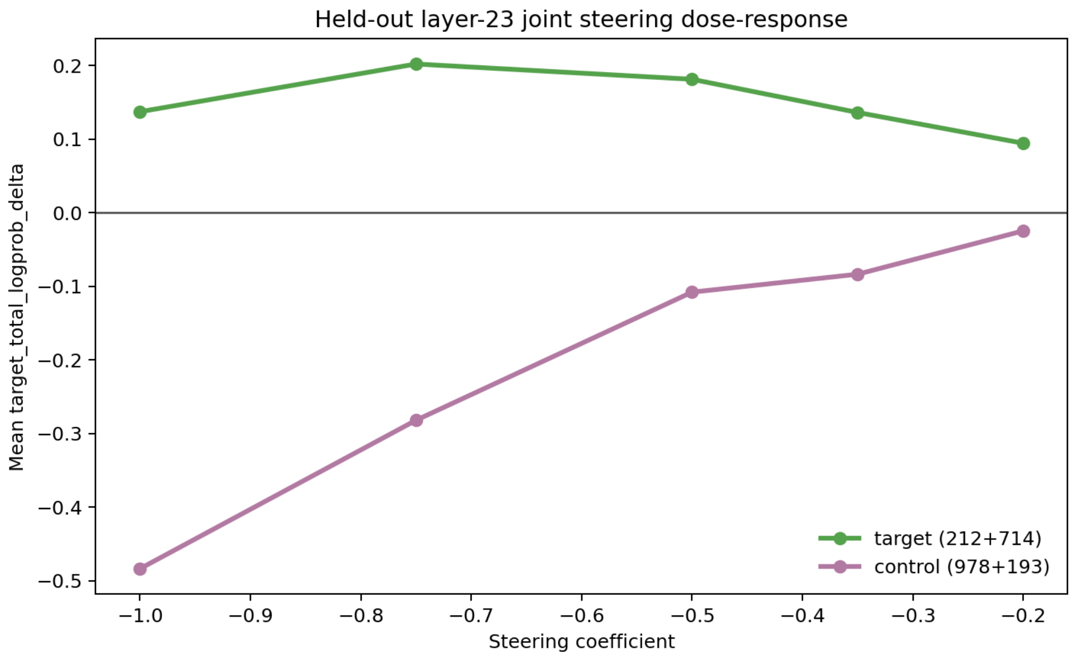

Figure 3: Held-out layer-23 joint steering dose-response. The target subspace $\{212, 714\}$ shows a positive dose-response, while the matched control subspace $\{978, 193\}$ does not reproduce the effect.

The target subspace exhibits a clean positive dose-response. The matched control subspace does not replicate the effect. At the best target dose,

$$\alpha^* = -0.75, \qquad \mathbb{E}[\Delta \log p] = +0.2019, \qquad \mathbb{E}[\mathrm{LR}] = 1.2431.$$

For the control set, the best tested coefficient still underperforms baseline:

$$\max_{\alpha} \mathbb{E}[\Delta \log p_{\text{control}}] = -0.0247.$$

This rejects the weak hypothesis that “any late-layer direction works” and instead supports a narrow, feature-specific causal account.

4 Single-feature and combo evidence

Before the held-out matched-control pass, single-feature steering identified features 212 and 714 as the most useful late-layer control handles. Joint steering outperformed either feature alone on the tested floodplain prompts.

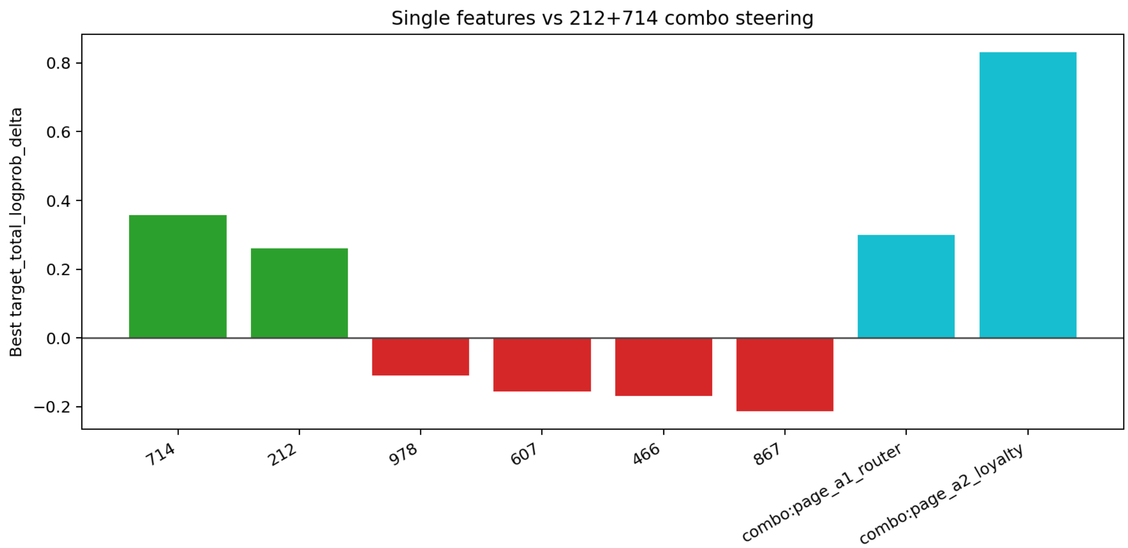

Figure 4: Best observed target-likelihood improvements for selected single features and the 212+714 combo from the available steering runs.

Best observed gains from the available steering runs:

- feature 212 at $\alpha = -0.5$: $\Delta \log p = +0.2597$
- feature 714 at $\alpha = -0.5$: $\Delta \log p = +0.3574$
- combo $(212 + 714)$ on the strongest tested prompt: $\Delta \log p = +0.8308$

5 Methods summary

The full industrial mechinterp paper draft can use the following compressed methods description:

1. capture layerwise activations from base and adapter modes;
2. train SAEs on captured activations;
3. compare dictionaries cautiously, preferring fixed-base encoding over raw feature-ID matching;
4. identify top drift features in the late active layer;
5. run causal steering sweeps on selected features and feature subspaces;
6. evaluate effects using authored-target likelihood rather than greedy text equality alone.

6 Limits of the current artifact set

The industrial mechinterp runs themselves were not included here because the `.npy` capture files dominate the folder size. As a result, this packet includes **causal and optimizer figures**, but not the layerwise drift localization plot from the larger industrial run. That drift result is still summarized in the paper wrap and should be included in the final paper once the industrial JSON or summary artifacts are added.

7 Future Work

The present results point toward a broader research program in which alignment is studied not only as output behavior, but as the preservation, reweighting, and controllable manipulation of latent geometric structure.

First, future work should test whether the late-layer feature subspace identified here generalizes beyond the current floodplain benchmark slice. The most immediate next step is held-out evaluation across additional worlds, prompts, and adapter variants to determine whether the same layer-localized mechanism recurs or whether different constitutional settings recruit distinct reasoning subspaces.

Second, the causal interventions used here should be strengthened. The current steering results show coefficient-sensitive movement in target likelihood, but not robust greedy output flips. More decisive causal tests should include multi-feature steering, feature ablation, reranking over candidate endings, and evaluator-based scoring. These approaches would help determine whether the identified late-layer features are merely correlated with constitutional reasoning or whether they function as reliable control handles over downstream decisions.

Third, future work should investigate whether optimizer choice affects the clarity and transportability of alignment-relevant latent structure. In our SAE benchmarks, Muon-hybrid improved reconstruction quality and reduced dead latents relative to AdamW, suggesting that geometry-aware optimization may produce cleaner feature structure. A natural next question is whether such optimizers better preserve or expose alignment-relevant subspaces during training and adaptation.

More broadly, this raises the possibility that *geometry isomorphism* may be important for scaling alignment: if useful alignment-relevant subspaces can be approximately preserved across models, layers, training stages, or adapter regimes, then alignment interventions may become more modular and interpretable. On this view, the goal is not only to enforce safety through global behavioral shaping, but to identify, preserve, and controllably reweight the latent geometries that underwrite desirable reasoning.

Finally, the industrial mechinterp pipeline introduced here should be extended to larger models, richer scenario corpora, and multiple adapter families. If late-layer feature reweighting proves to be a recurring mechanism, then sparse feature geometry may provide a scalable bridge between mechanistic interpretability and alignment engineering.

8 Conclusion

This paper asked a simple mechanistic question: where does a constitutional adapter actually act inside a small language model? Across balanced-world SAE analysis, fixed-base drift measurement, and held-out causal steering, the answer in this setting is consistent: the adapter's measurable effect is concentrated in a late transformer layer, where it reweights a narrow set of reasoning-relevant features rather than broadly rewriting the network.

This result matters both methodologically and empirically. Methodologically, it shows that adapter mechanistic interpretability benefits from industrialized analysis pipelines and from fixed-dictionary comparisons rather than naive alignment of independently trained sparse autoencoders. Empirically, it shows that late-layer latent structure is not only descriptively shifted by the adapter, but causally connected to authored-target likelihood under feature steering.

The broader implication is that scalable alignment may depend less on globally reshaping model behavior and more on identifying and controlling a relatively small set of alignment-relevant latent geometries. If those geometries can be preserved, compared, and manipulated across training regimes and model scales, then alignment interventions may become more modular and interpretable.

In this sense, the present work is less a final answer than a starting point. It provides initial evidence that constitutional adaptation can be localized, measured, and partially controlled in feature space, and that latent feature geometry may offer a promising foundation for future alignment research.